PROJECT ZERO
VERSION 1

DOMINUS

POWERED BY PI , CONTROLLED WITH PRECISION

CREATED AND DESGINED BY
ANUBHAV SINGH

About the Author

Anubhav Singh is an engineering enthusiast focused on building practical hardware systems that combine computing, electronics, and network infrastructure. His work explores self-hosted technologies, remote system management, and modular hardware design.

PROJECT-0 represents an experimental platform developed to study controlled computing environments where power management, system switching, and remote accessibility are integrated into a single architecture.

Through hands-on experimentation and iterative design, the project reflects an

Acknowledgement

The development of PROJECT-0 would not have been possible without the support and encouragement of several individuals and resources.

Special thanks to my mentors, colleagues, and the broader open-source community whose knowledge, documentation, and shared ideas continue to inspire experimentation and learning.

I would also like to acknowledge the countless engineers and developers whose work in hardware, software, and networking technologies made it possible to explore and build systems like this.

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1 INTRODUCTION

WHAT EXACTLY IS PROJECT-0?

PROJECT-0 is a multi-system computing architecture controlled by a Raspberry Pi through precise voltage regulation, relay logic, and dual-mode operation (web + manual).

In simple words:
It is a two-computer system that behaves like one intelligent, remotely managed unit.

A low-power controller stays Online 24/7.
High-power systems activate only when required.

Control happens through:
Web interface (remote logic control)
Manual DPDT switching (hardware override)
Isolated relay architecture
Controlled disk selection for multi-OS deployment
It separates logic from power, and control from execution.

Not just a cabinet.
Not just two PCs.
A controlled computing environment.

WHY WAS PROJECT-0 BUILT?

Switching hardware manually is frustrating.
Changing disks physically is inefficient.
Running a full server 24/7 wastes power.
Cloud services charge monthly.
And relying on external platforms means your data lives somewhere else.

PROJECT-0 was built to solve that.

Instead of:
Keeping a heavy system running all day
Paying for cloud storage
Constantly rewiring disks
Depending on third-party infrastructure

The idea was simple:
Keep only a low-power controller Online,
and activate full compute systems only when required —
from anywhere in the world.

WHAT CAN PROJECT-0 ACTUALLY DO?

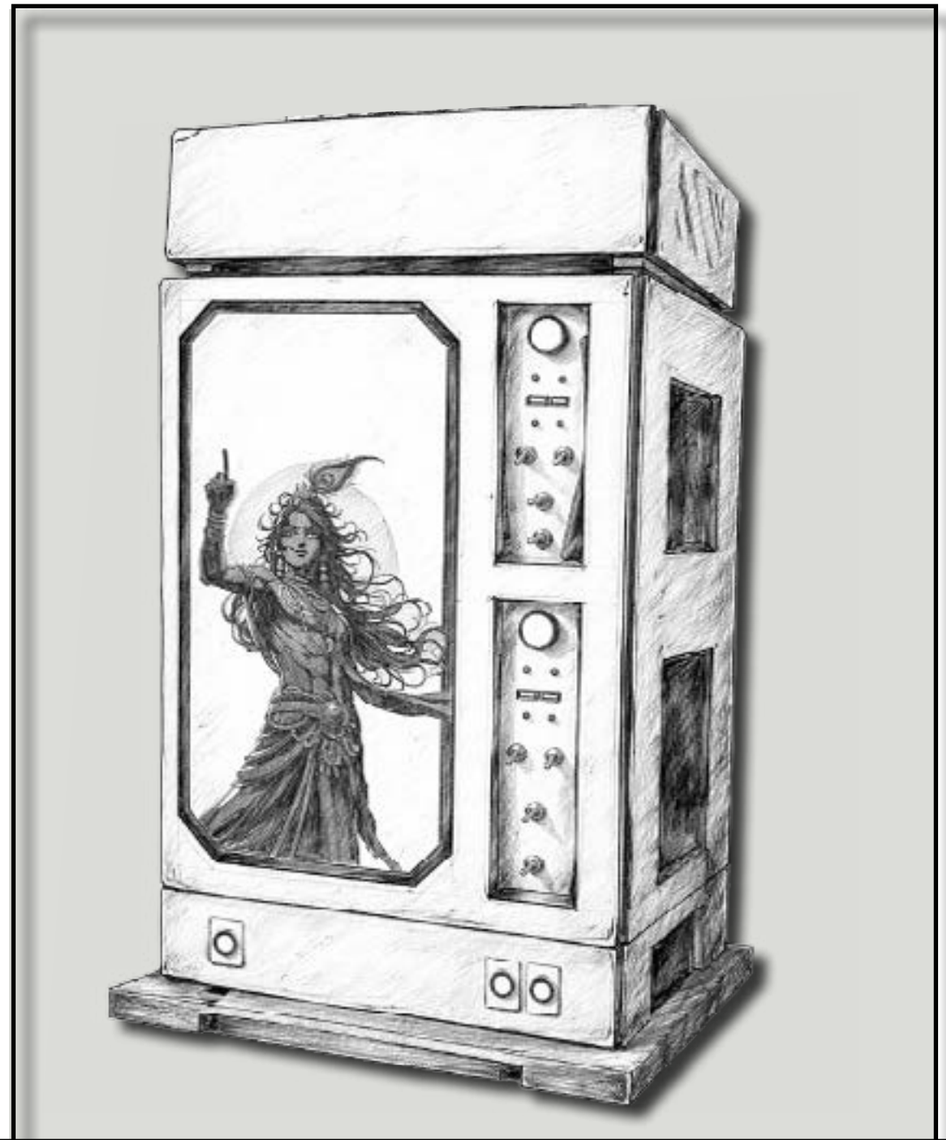
At its core, PROJECT-0 functions as a self-hosted home lab server infrastructure.

It combines:
Multi-OS compute environment
Remote power management
Disk-level selection control
Secure remote access (VPN + domain routing)
BIOS-level recovery via KVM

It can operate as:
A personal cloud server
A home data server
A remote AI agent system
A private storage infrastructure

Instead of uploading your data to external cloud providers,
you host your own.
It effectively replaces many common cloud services
with a private, locally owned architecture.

2 HARDWARE LAYOUT OVERVIEW OF PROJECT-0



TOP-SIDE VIEW

LEFT-SIDE VIEW

FRONT-SIDE VIEW

BACK-SIDE VIEW

RIGHT-SIDE VIEW

Controller Base

The controller unit acts as the central management layer of PROJECT-0. It contains the Raspberry Pi control processor along with relay switching modules, optocoupler isolation circuits, and voltage regulation units. This layer manages system activation, disk selection logic, and safe electrical isolation between control signals and high-power switching components.

Raspberry Pi 3B+ — Control Processor
The Raspberry Pi functions as the primary control processor of PROJECT-0. It executes the control logic, manages GPIO signaling, hosts the web interface, and coordinates relay switching for system power and disk selection.

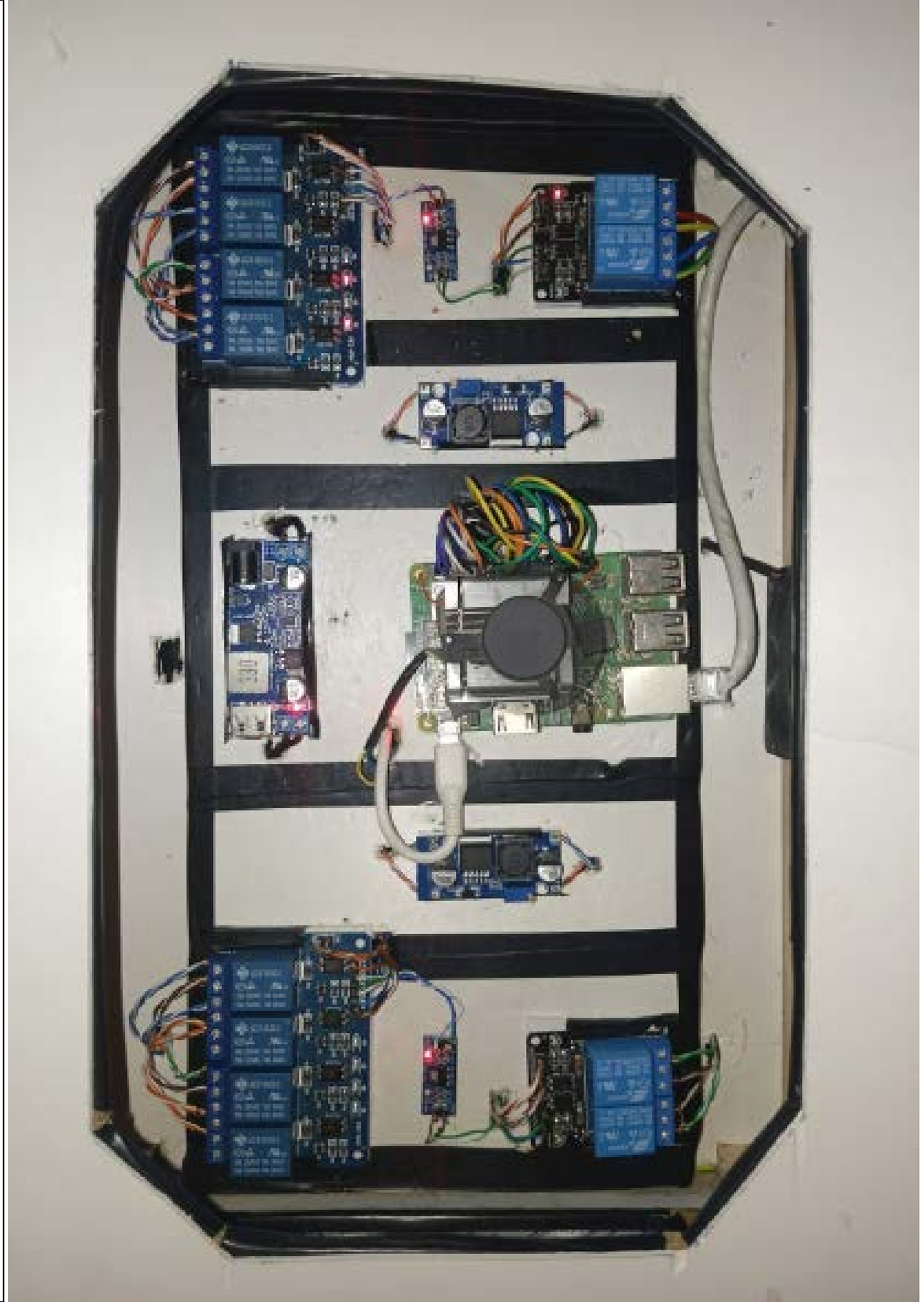
12V to 5V Converter — Relay & Controller Power Supply
This high-current DC–DC converter steps down the 12V supply to a stable 5V rail used to power the relay modules and controller cooling fans. It provides sufficient current capacity (up to 5A) to ensure reliable operation of the control electronics and auxiliary components.

Relay Modules — Power Switching Interface
Relay modules act as electrically controlled switches that allow the Raspberry Pi to manage high-power circuits safely. In PROJECT-0 they control the 230V supply to System-1 and System-2, motherboard power-on signals, and selected 12V/5V control lines used for disk and system activation.

5V to 3.3V Converter — Logic Level Supply
This module converts 5V supply to a stable 3.3V logic level required by the Raspberry Pi GPIO interface. It ensures safe signal compatibility with the relay control circuits while maintaining a shared common ground reference with the controller.

Optocoupler Modules — Electrical Isolation
Optocouplers isolate the Raspberry Pi's low-voltage GPIO circuitry from relay switching circuits. This prevents voltage spikes or noise from reaching the controller.

Buck Converters — Voltage Regulation
DC-DC buck converters regulate incoming voltage to stable levels required by the controller and relay modules, ensuring reliable operation of all control electronics.



System-1 Unit

System-1 acts as the primary high-performance computing node of PROJECT-0. This unit contains the main processing hardware, graphics processing unit, power supply, manual control switches, and supporting network and cooling components required for stable system operation.

Compute Platform — Motherboard, CPU, RAM & GPU
System-1 is built on an H81 motherboard equipped with an Intel Core i7-4770 processor and 16GB DDR3 1600 MHz RAM, forming the primary computing platform. The system also integrates an NVIDIA GTX 1650 graphics card to support GPU-accelerated workloads and smooth remote streaming performance.

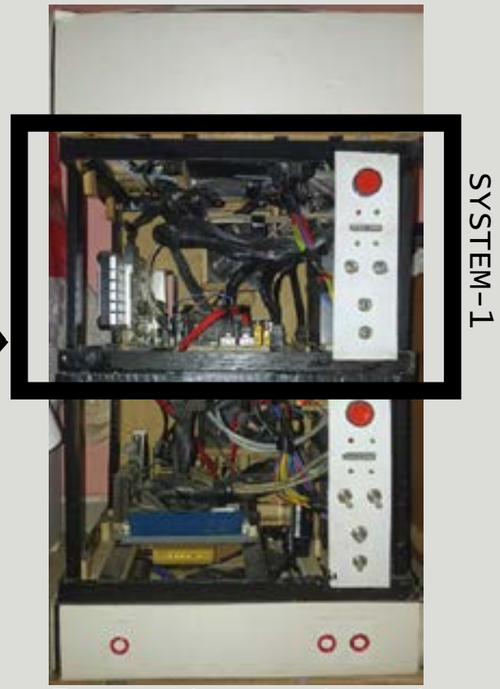
SMPS — System Power Supply
An 80+ Bronze certified power supply converts 230V AC input into regulated DC power required by the motherboard, GPU, storage devices, and peripherals, ensuring stable operation during continuous 24/7 usage.

SSD Storage — Operating System Drives
System-1 uses two SSD drives for operating system deployment. Disk-1 (256GB SSD) is assigned for Windows, while Disk-2 (128GB SSD) is assigned for Linux. Disk selection is controlled through relay logic and

DPDT Switches — Manual Control Interface
DPDT switches provide hardware-level control for disk selection and operation modes, allowing the system to operate manually when the controller is unavailable.

Optocoupler Modules — Electrical Isolation
Optocouplers isolate control signals between the controller and the system hardware, protecting components from electrical noise while also enabling the controller to detect manual disk selection states.

Network & Cooling — Connectivity and Thermal Control
CAT6 Ethernet cables and connectors provide reliable network communication between the system and the controller, while cooling fans maintain airflow to regulate internal temperature during extended workloads.



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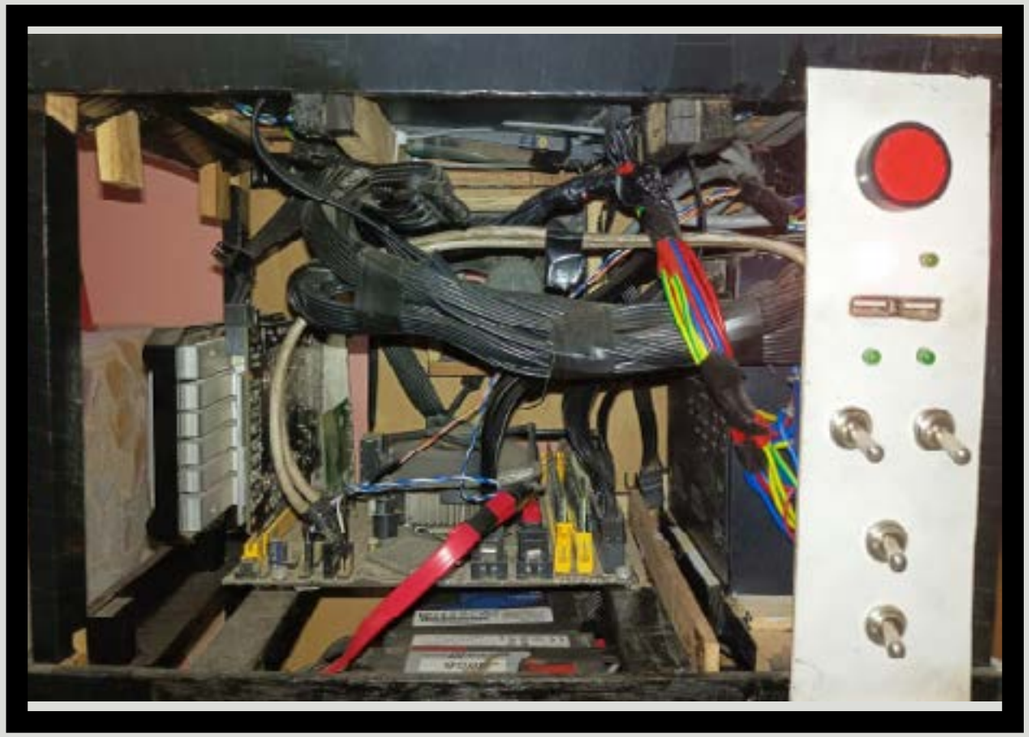

System-2 Unit

System-2 functions as the secondary computing node of PROJECT-0, designed to support server operations and auxiliary processing tasks. This unit contains the core compute hardware, storage devices, power supply, manual control switches, and supporting network and cooling components required for reliable system operation. It enables dedicated services such as home-server workloads and the AI agent environment while remaining fully integrated with the controller-based management system.

Compute Platform — Motherboard, CPU, RAM & GPU
System-2 is built on an H81 motherboard equipped with an Intel Core i5-4470 processor and 16GB DDR3 1600 MHz RAM, forming the secondary compute platform of PROJECT-0. The system also integrates an NVIDIA GTX 210 graphics card, providing basic graphical output and remote desktop capability.

SMPS — System Power Supply
An 80+ Bronze certified power supply converts 230V AC input into regulated DC power required by the motherboard, GPU, storage devices, and peripherals, ensuring stable operation during continuous 24/7 usage.

SSD Storage — Operating System Drives
System-2 uses multiple storage devices to support different system roles. Disk-1 consists of a 128GB SSD paired with a 2TB HDD for server storage and data services, while Disk-2 uses a 30GB SSD dedicated for running

DPDT Switches — Manual Control Interface
DPDT switches provide hardware-level control for disk selection and operation modes, allowing the system to operate manually when the controller is unavailable.

Optocoupler Modules — Electrical Isolation
Optocouplers isolate control signals between the controller and the system hardware, protecting components from electrical noise while also enabling the controller to detect manual disk selection states.

Network & Cooling — Connectivity and Thermal Control
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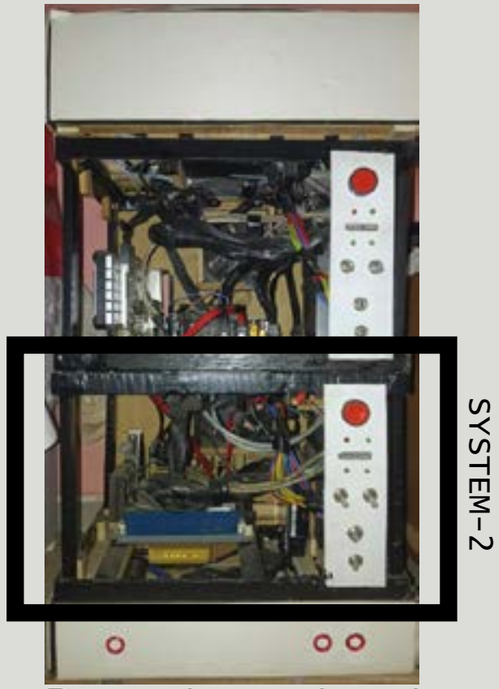

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Power Base

The power base forms the electrical and network distribution layer of PROJECT-0. It manages AC input power, system protection, controller power supply, and network connectivity for both computing units and the controller system.

AC Power Input & Protection
The base includes the main 230V power socket, fuse holder, and protection fuse which provide safe electrical input to the system. This stage protects the hardware from overcurrent conditions and distributes AC power to the system power supplies and controller adapters.

Power Supply Modules
Dedicated 12V and 5V power adapters provide regulated power for the controller electronics and supporting modules. These supplies ensure stable voltage delivery for the Raspberry Pi controller and relay control circuitry.

System Status Indicators
230V indicator LEDs are used to visually display system power status. Separate indicators show the main power state and the active power status of System-1 and System-2, allowing quick operational monitoring.the

DPDT Switches — Manual Control Interface
DPDT switches provide hardware-level control for disk selection and operation modes, allowing the system to operate manually when the controller is unavailable.

Network Distribution
A 5-port Ethernet switch, along with CAT6 cables and connectors, provides the local network backbone of PROJECT-0. This allows communication between the controller, System-1, System-2, and the external router

TOP-SIDE VIEW

BACK-SIDE VIEW

CONTROLER BASE

SR.NO	Component Name	Qty
1	Raspberry pi 3b+	1
2	64GB SD card	1
3	12v to 5v Convetor	1
4	Buck convotor	2
5	5v to 3.3v Convotor	2
6	12v fan	2
7	2-channel relay	2
8	4-Channel relay	2
9	2-Channel octocopler	3

SYSTEM 1 BASE

SR.NO	Component Name	Qty
1	H81 motherboard	1
2	I7 4790 processor	1
3	8GB DDR3 ram	2
4	450W Power Supply	1
5	Msi GTX 1650 Graphic Card	1
6	Ethernet cable	4
7	256GB SSD for DISK1	1
8	128 GB SSD for DISK2	1
9	6-pin DPDT switch	4
10	Push-Button	1
11	12v Fan	2

SYSTEM 2 BASE

SR.NO	Component Name	Qty
1	H81 motherboard	1
2	I5 4590 processor	1
3	8GB DDR3 ram	2
4	550W Power Supply	1
5	Gtx 210 Graphic Card	1
6	Ethernet cable	4
7	2TB HDD for Disk1	1
8	128GB SSD for Disk1	1
9	30GB SSD for Disk2	1
10	6-pin DPDT switch	4
11	Push-Button	1

POWER BASE

Sr.No	Component Name	Qty
1	5-pin socket	3
2	230V led	3
3	Fuse and Fuse Socket	2
4	3-pin DPDT switch	2
5	230V to 12v 5amp Adapter	1
6	230V to 5V 2amp Adapter	1
7	5-Port Ethernet Switch	1
8	thernet cable and connectc	1

3 SYSTEM LOGIC & ARCHITECTURE

SO IF YOU ARE IN A REMOTE-MODE

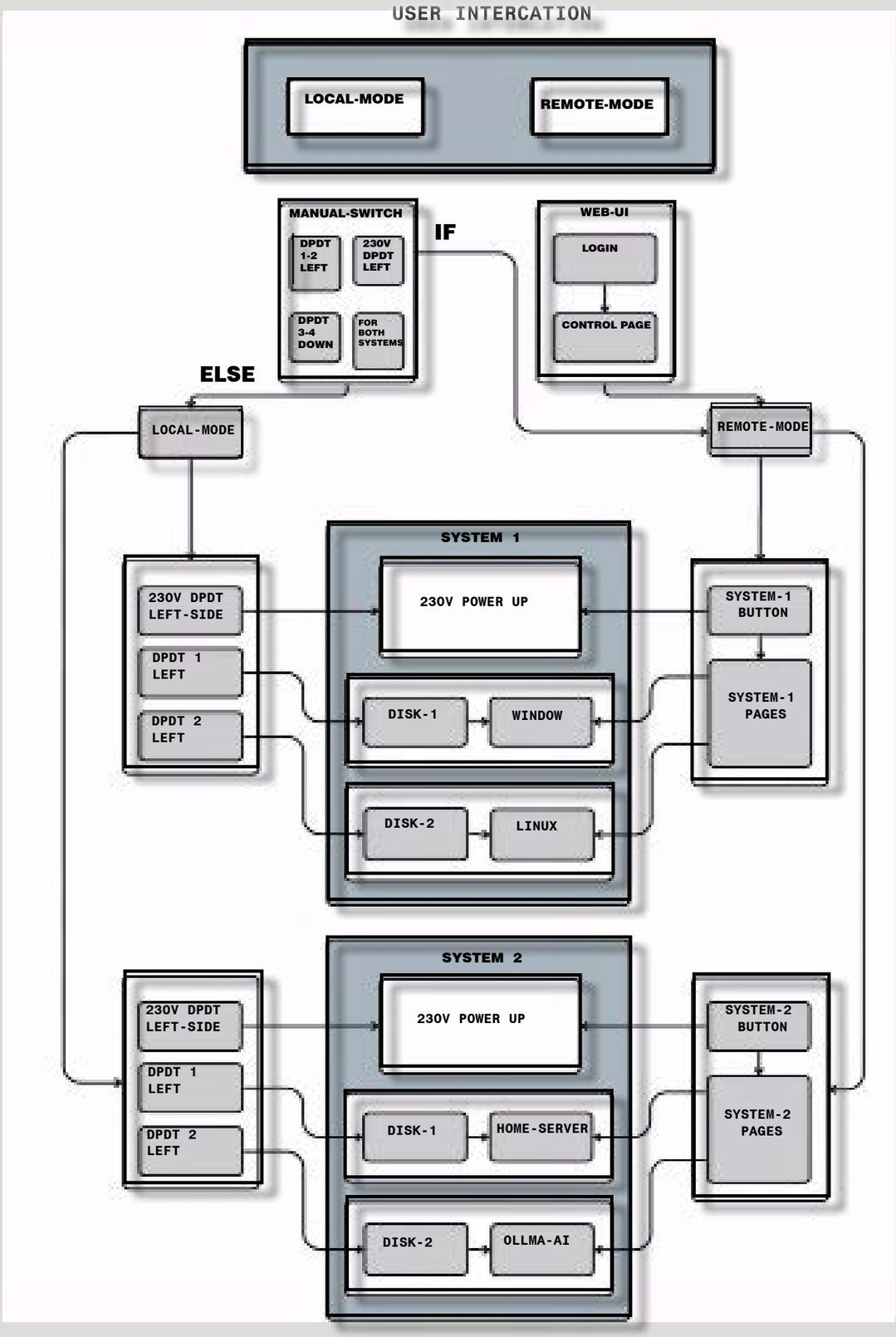
Simple.
Before anything, shift the hardware to remote configuration.

Set DPDT 1–2 (both systems) to LEFT.
Set DPDT 3–4 to UP (Remote Mode).
Now the relay logic, disk selection, and power triggers are fully governed by the Raspberry Pi controller.
Manual override is isolated.
At this point, the controller takes authority.
Open your browser ---> Enter the local IP or your domain ---> Login → Go to the Control Page.
Select:- System 1 or System 2 -->Disk 1 or Disk 2 (based on required OS) --->Then press Power ON.
The controller now handles everything internally — relay sequencing, disk routing, motherboard power trigger, and sensing pin feedback. Disk switching is electrically isolated to prevent conflict. Power logic follows the defined sequence to avoid surge or boot inconsistency.
Status updates reflect in real time.
No manual toggling. No cross-wiring risk

WHAT IF CONTROLLER FAIL OR SOME HARDWARE ISSUE ON CONTROLLER SIDE

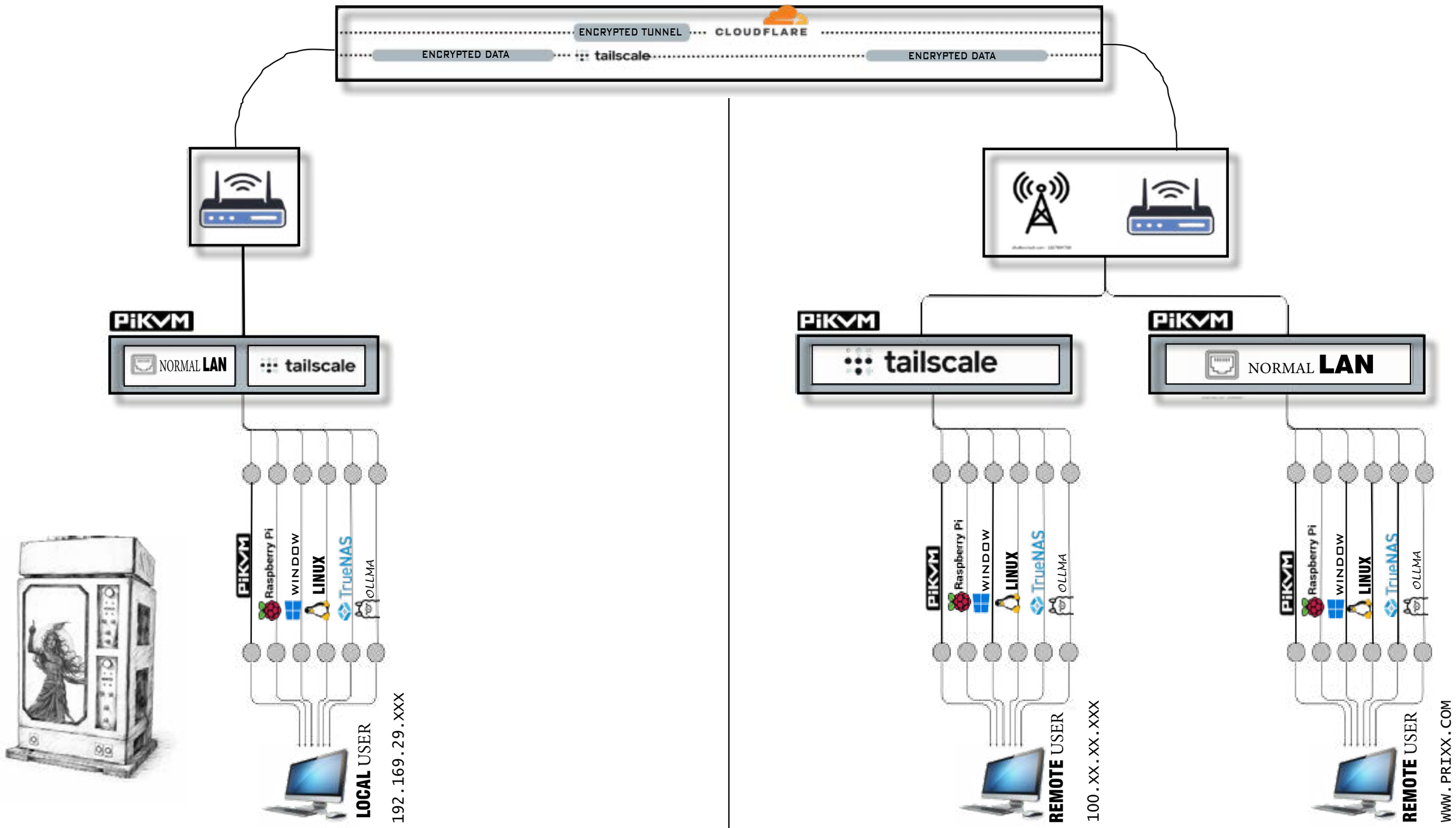
No panic. It built for this.

If remote logic fails or the controller becomes unavailable, switch back to Manual Mode.
Set DPDT 3–4 to DOWN (Manual Mode).
Set the 230V DPDT to LEFT (direct supply path).
At this point, control shifts fully to hardware. The controller is electrically bypassed. Power and disk routing are now governed only by mechanical switching.
Here, precision matters.
Suppose For Disk 1 selection:
Set DPDT 1 → RIGHT --> Set DPDT 2 → MIDDLE (neutral — no power on this line)
Never power both lines simultaneously. The middle position physically isolates the unused disk, preventing cross-power conditions or back feed. This is critical for disk safety and motherboard stability.
Apply the same logic to the second system.
Now you are running purely mechanical control, No web interface, No controller logic, No software dependency.
Worst case?
Still operational.



NETWORK ARCHITECTURE

Network architecture defines how devices within the system communicate and exchange data. In PROJECT-0, the router and Ethernet switch form the local network backbone, connecting the Raspberry Pi controller, System 1, and System 2. Secure remote access is achieved through VPN tunneling and optional domain routing, enabling controlled system management from both local and external networks.



WHAT IF !

SO IF YOU ARE IN A HOME-ENVIRONMENT

You are sitting in front of the system, same room, same network.

Just Open your browser.
Type the controller's local IP 192.168.29.xx
Login to Raspberry pi web and login to Control Page

Pick the system 1 and system 2
Select Disk 1 or Disk 2 as per use, depending upon what OS you need to use after that System ON, all things were updated through the sensing pin in real-time.

Then if you use window/Linux open moonlight as in DOMIUS side sunshine is installed and you good to go
For using servers and agent it can login through the web directly using local ip 192.168.29,XX

WHAT IF YOU ARE IN A OUTSIDE-ENVIRONMENT

You're not local. Maybe another city. Maybe another country.

No problem.

Open your browser.
Instead of the local IP, type:- www.primxx.com

The domain routes securely to your controller.
Encrypted connection. Same control. Same response. Just remote.

Login → Go to Control Page → Select System 1 or System 2 → Select Disk 1 or Disk 2 as required → Power ON.

Once powered, open Moonlight.
Now here inssert the assign domain and you good to go and now you have full access — Windows or Linux — exactly same

For Server and agent use the asgin domin in web like htttp://www.prxx and you have all the acces

WHAT IF YOU ARE IN A OUTSIDE-ENVIROMENT PLUS YOUR DOMAIN GET EXPIRED

Okay. Worst case.
Domain is gone. DNS is dead. Website not reachable.
Still no panic.

Open Tailscale.
Connect to your private network.
Every device is already part of the mesh.
Use the controller's Tailscale IP instead 100.xx.xx.xxx of the public domain.
Access the web interface directly through that secure tunnel.

Login → Control → Select system → Select disk → Power ON.
Nothing changes in the core architecture — only the access path changes.

Then use the system's Tailscale IP inside Moonlight.
Sunshine detects the host over the private network.
Connect.

Direct. Encrypted. No public dependency.
Domain is optional.
Architecture is not.

WHAT IF OS CRASH OR SOME BIOS LEVEL ISSUE ON ANY SYSTEMS

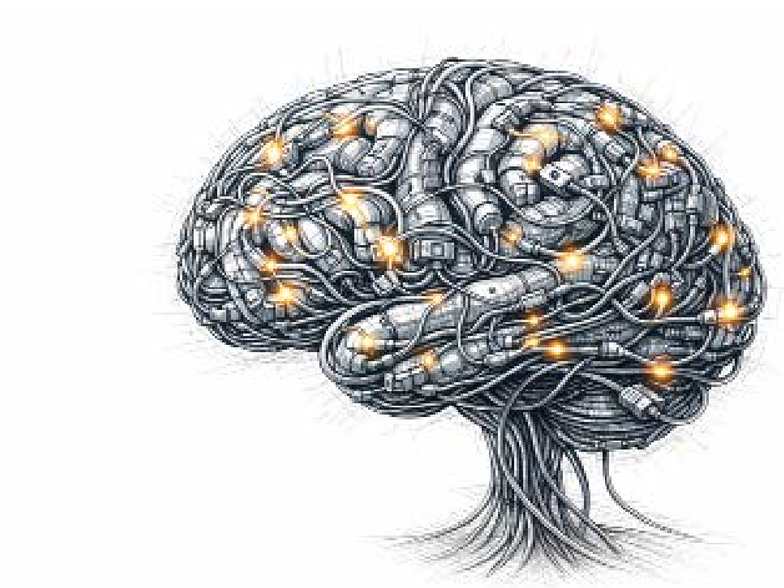
Blue screen. Black screen. No display. No response.

If you are at home, sitting in front of the machine, this is not a serious issue. You can connect a monitor, plug in a keyboard, and fix it manually. The real problem begins when you are outside — another office, another city, another country — and the system stops responding with no physical access available.

Still no problem. Switch the controller to KVM mode, select System 1 or System 2, and access it through PiKVM. Now you are not inside the OS; you are at BIOS level, with full hardware-level keyboard, video, and mouse control remotely.

From there you can repair the OS, reinstall it, check hardware, modify BIOS settings, change boot order, mount an ISO remotely, and recover everything. Even if the disk is corrupted or the operating system is completely dead, access remains intact.

4 OPERATION & EVALUATION



"A well-designed system is not the one with the most wires, but the one where every wire has a purpose."

OPERATING MODES

PROJECT-0 operates in two primary modes:

Remote Mode for network-based control, and

Local Mode for direct hardware interaction.

This dual-mode design ensures the system remains accessible, flexible, and operational under all conditions.

WHY DOES PROJECT-0 USE TWO OPERATING MODES?

PROJECT-0 is designed with a dual-mode architecture to ensure continuous system availability, flexible access, and fault tolerance under different operating conditions. The system operates in Remote Mode and Local Mode, each serving a specific purpose within the overall control architecture.

Remote Mode is the primary method of operation, allowing the user to access and control the system from anywhere through a web-based interface. In this mode, the Raspberry Pi controller manages relay switching, disk selection, and system power states. This enables remote system activation, multi-OS selection, and full control without physical interaction, making the system function like a self-hosted remote computing platform.

However, relying entirely on software control introduces a single point of failure. To address this, Local Mode is implemented as a hardware-level fallback mechanism. In this mode, the system bypasses the controller and operates through direct DPDT switch control. This allows manual selection of disks and system activation even if the controller fails, the network is unavailable, or remote access is not possible.

Local Mode also provides immediate physical control when the user is present on-site, eliminating dependency on network latency or software interfaces.

The dual-mode design ensures that the system is:

Remotely accessible when connected
Locally controllable when required
Fail-safe in case of controller or network failure

This architecture combines automation with manual redundancy, ensuring that the system remains operational under all conditions.

REMOTE MODE OPERATION

Remote Mode allows full system control through the web interface, enabling access to PROJECT-0 from anywhere within the network or through remote connectivity.

To activate Remote Mode, set the hardware switches to controller control: DPDT 1–2 to the left position and DPDT 3–4 to the upper position (as shown in the adjacent diagram). This configuration transfers system control to the Raspberry Pi controller.

Once configured, open a web browser and enter the controller address:
192.168.29.118:5000

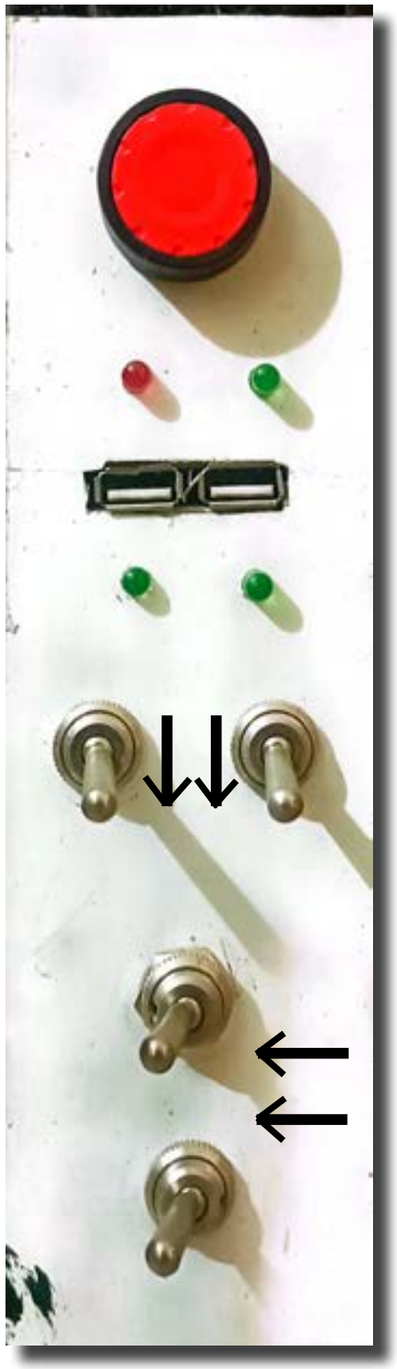
After login, the dashboard displays the current system status using sensing inputs. If a system shows “Online”, it indicates that the system is already powered ON or currently in use.

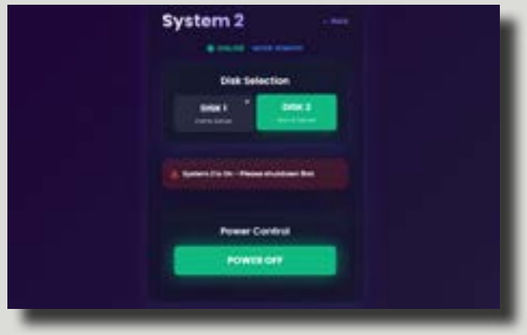
From the dashboard, select the required system (System-1 or System-2) and choose the desired disk (Disk-1 or Disk-2) based on the operating system requirement. After selection, trigger the power control from the interface to start the system.

The controller automatically manages relay switching, disk activation, and safe startup sequencing, allowing the system to operate without any physical interaction.

Note

Ensure correct DPDT switch positions before accessing the system
If status shows Online, avoid re-triggering power
Use remote mode only when controller and network are stable





LOCAL MODE OPERATION

Local Mode allows direct hardware control of PROJECT-0 without relying on the controller or network. This mode is used when operating locally, or in scenarios where the controller or remote access is unavailable.

To activate Local Mode, set DPDT 3–4 to the downward position (as shown in the diagram). This bypasses the controller and transfers full control to the hardware switches.

Disk selection is performed manually:

For Disk-1, set DPDT 1 to the right position

For Disk-2, set DPDT 2 to the right position

After selecting the required disk, press the red push button to power ON the system. The same procedure applies to both System-1 and System-2

Note

Local Mode requires precise switch handling. Incorrect configuration may lead to hardware conflicts or system failure.

Never activate both disks at the same time
Always keep the unused disk in a neutral (OFF) position
Verify switch positions before pressing the power button
Do not change disk selection while the system is running
Ensure the system is completely OFF before switching disks

ADVANTAGES

PROJECT-0 enables a self-hosted computing environment that reduces dependency on external services and traditional always-on systems. By integrating a controller-based architecture with on-demand system activation, it significantly lowers power consumption while maintaining full computing capability when required.

The system can function as a home server platform, allowing storage, applications, and services to be hosted locally instead of relying on third-party cloud providers. This reduces the need for external platforms such as common cloud-based applications, while maintaining full control over data and system behavior.

With remote access capability, the system allows a full desktop environment to be accessed from anywhere using only a display device, input peripherals, and an internet connection, effectively reducing the need for a dedicated high-performance computer at every location.

Additionally, the combination of remote control and manual fallback ensures flexibility, reliability, and continuous usability under different operating conditions.

LIMITATIONS

The primary limitation of the system is its dependency on continuous power availability. In the event of a power outage, system access and operation are interrupted unless supported by an external backup power solution.

The system also relies on stable network connectivity for remote operation. Without internet access, remote mode becomes unavailable, and the system must be operated in local mode.

Furthermore, the architecture requires careful manual handling in local mode, and improper switch configuration may lead to operational issues. The system is designed for controlled usage and may require technical understanding during setup and operation.

"First make it work.
Then make it beautiful.
Then make it fast."

Looking Ahead

PROJECT-0 is only the beginning.
The next version will be more compact, faster, more precise, and easier to use — built from everything learned in this version.